

DSA0603 Data Handling and Visualization

Slot B

CO1 – Assessment Test 3 (AT3)

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Question 1 – Integrated Retail Visualization

```
library(ggplot2)
```

```
library(scales)
```

```
retail <- data.frame(
```

```
  Branch=c("B1","B2","B3","B4","B5","B6","B7","B8","B9","B10",  
           "B11","B12","B13","B14","B15","B16","B17","B18","B19","B20"),
```

```
  Region=c("North","South","East","West","North","South","East","West",  
           "North","South","East","West","North","South","East","West",  
           "North","South","East","West"),
```

```
  Category=c("Electronics","Furniture","Clothing","Food","Electronics",  
             "Furniture","Clothing","Food","Electronics","Furniture",  
             "Clothing","Food","Electronics","Furniture","Clothing",  
             "Food","Electronics","Furniture","Clothing","Food"),
```

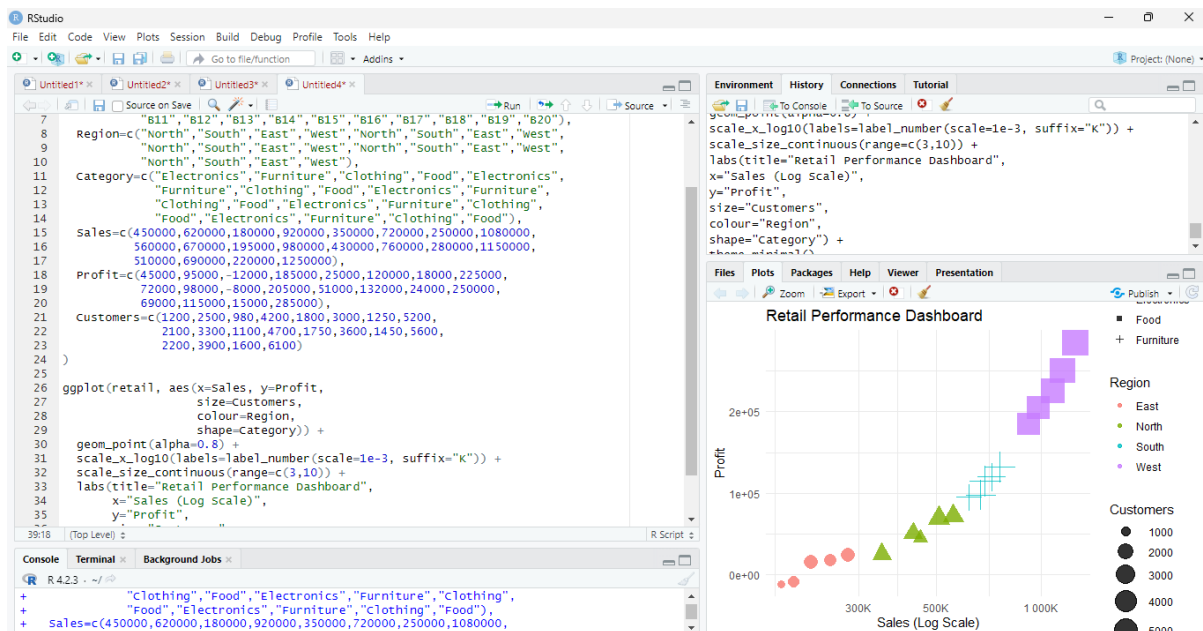
```
  Sales=c(450000,620000,180000,920000,350000,720000,250000,1080000,  
          560000,670000,195000,980000,430000,760000,280000,1150000,  
          510000,690000,220000,1250000),
```

```
  Profit=c(45000,95000,-12000,185000,25000,120000,18000,225000,  
          72000,98000,-8000,205000,51000,132000,24000,250000,  
          69000,115000,15000,285000),
```

```
  Customers=c(1200,2500,980,4200,1800,3000,1250,5200,  
              2100,3300,1100,4700,1750,3600,1450,5600,  
              2200,3900,1600,6100)
```

```
)
```

```
ggplot(etail, aes(x=Sales, y=Profit,
                 size=Customers,
                 colour=Region,
                 shape=Category)) +
  geom_point(alpha=0.8) +
  scale_x_log10(labels=label_number(scale=1e-3, suffix="K")) +
  scale_size_continuous(range=c(3,10)) +
  labs(title="Retail Performance Dashboard",
       x="Sales (Log Scale)",
       y="Profit",
       size="Customers",
       colour="Region",
       shape="Category") +
  theme_minimal()
```



Interpretation and Justification

- **X-axis = Sales** and **Y-axis = Profit** show revenue and profitability simultaneously.
- **Point size = Customers** represents customer base magnitude.

- **Colour = Region** distinguishes geographical performance.
- **Shape = Category** differentiates product groups.

Scale Transformation

- Sales values range from **180,000 to 1,250,000**, so a **log10 scale** reduces skewness and prevents high-sales branches from dominating the plot.

Key Insights

- West region branches generally achieve the highest sales and profits.
- Clothing category includes negative-profit branches, indicating weaker performance.
- Larger customer bases are associated with higher sales in most regions.

Question 2 – Cartesian vs Polar Coordinates

```
library(ggplot2)
```

```
library(tidyr)
```

```
# Create the data frame
```

```
energy <- data.frame(
```

```
  Month = month.abb,
```

```
  Coal = c(620, 580, 590, 605, 640, 670, 690, 720, 700, 680, 650, 630),
```

```
  Gas = c(310, 295, 305, 320, 340, 355, 360, 370, 365, 350, 335, 320),
```

```
  Solar = c(45, 60, 90, 130, 180, 230, 250, 245, 210, 160, 90, 50),
```

```
  Wind = c(80, 95, 120, 150, 170, 185, 195, 190, 175, 150, 110, 90),
```

```
  Hydro = c(140, 150, 160, 170, 180, 195, 205, 210, 200, 185, 165, 150)
```

```
)
```

```
# Reshape data to long format
```

```
energy_long <- pivot_longer(
```

```
  data = energy,
```

```
  cols = -Month,
```

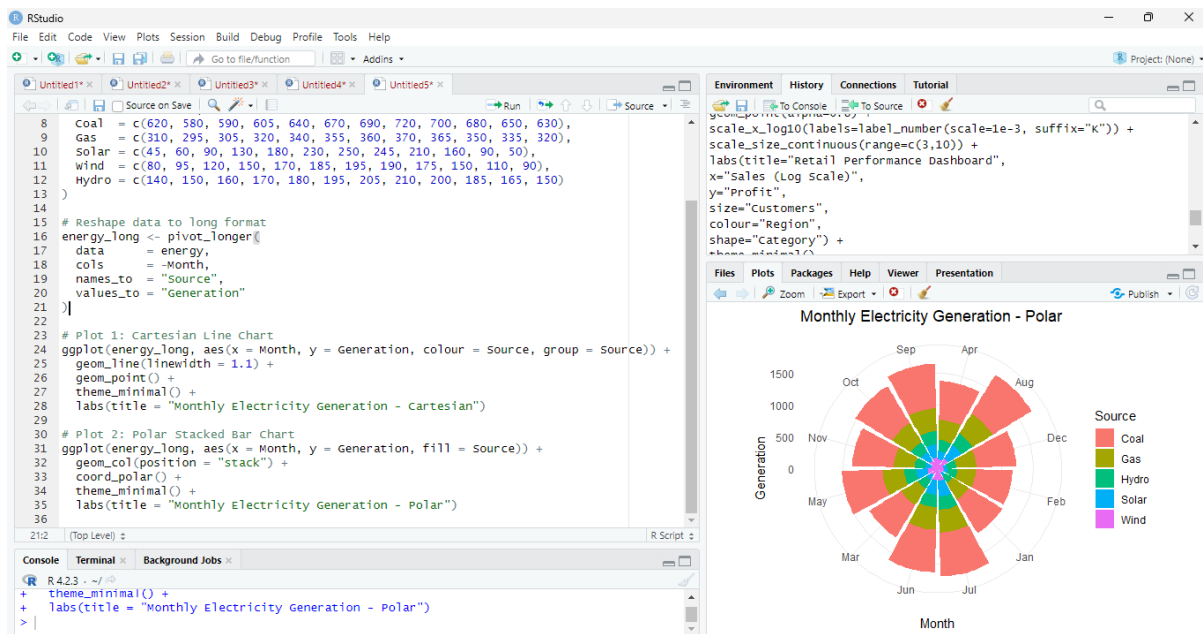
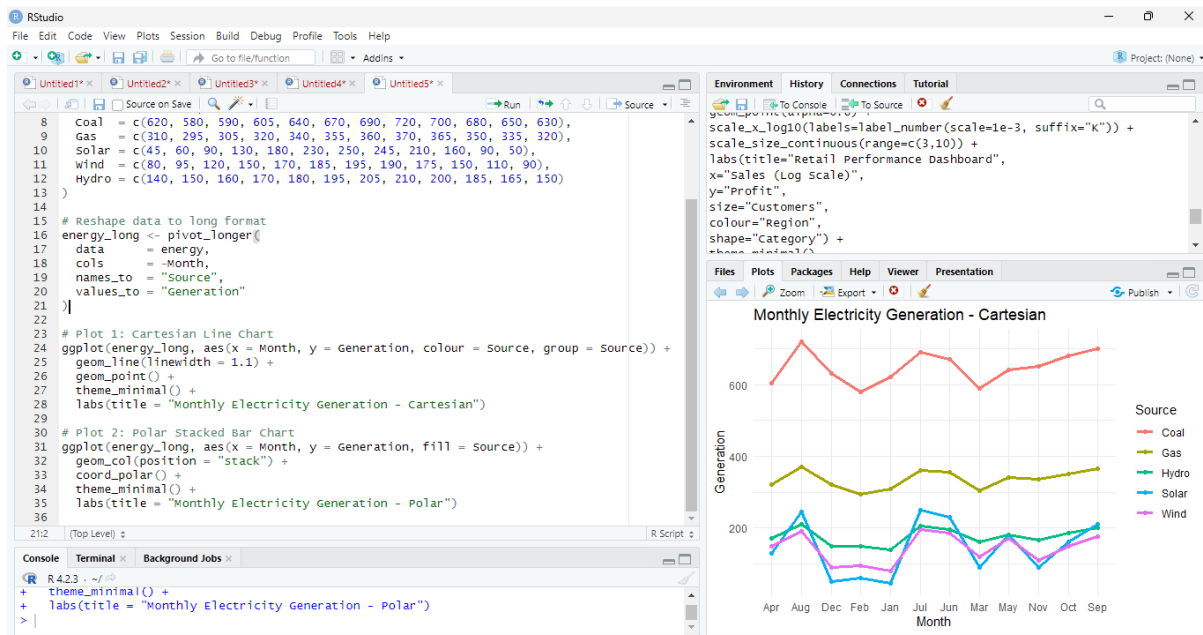
```
names_to = "Source",  
values_to = "Generation"  
)
```

Plot 1: Cartesian Line Chart

```
ggplot(energy_long, aes(x = Month, y = Generation, colour = Source, group =  
Source)) +  
  
  geom_line(linewidth = 1.1) +  
  
  geom_point() +  
  
  theme_minimal() +  
  
  labs(title = "Monthly Electricity Generation - Cartesian")
```

Plot 2: Polar Stacked Bar Chart

```
ggplot(energy_long, aes(x = Month, y = Generation, fill = Source)) +  
  
  geom_col(position = "stack") +  
  
  coord_polar() +  
  
  theme_minimal() +  
  
  labs(title = "Monthly Electricity Generation - Polar")
```



Comparison

Cartesian Coordinates

- Best for comparing exact monthly values.
- Clearly shows trends, peaks, and seasonal changes.

Polar Coordinates

- Emphasizes cyclical annual patterns.
- Useful for communicating seasonality but not precise comparisons.

Better Analytical Insight

Cartesian coordinates provide better analytical insight because line charts preserve a common baseline and support accurate month-to-month comparison

Question 3 – Climate Visualization with Colour Scales

```
library(ggplot2)
```

```
library(scales)
```

```
# Create the climate data frame
```

```
climate <- data.frame(
```

```
  City      = c("Delhi", "Mumbai", "Chennai", "Kolkata", "Hyderabad",  
                "Bangalore", "Pune", "Ahmedabad", "Jaipur", "Lucknow",  
                "Bhopal", "Patna", "Goa", "Kochi", "Nagpur", "Indore",  
                "Surat", "Visakhapatnam", "Coimbatore", "Mysuru"),
```

```
  Temperature = c(44, 35, 38, 40, 39, 31, 33, 43, 45, 39,  
                  38, 40, 32, 30, 41, 37, 36, 34, 29, 28),
```

```
  Humidity    = c(22, 84, 78, 72, 48, 66, 58, 26, 18, 42,  
                  36, 45, 89, 91, 30, 34, 76, 82, 68, 64),
```

```
  Rainfall    = c(6, 280, 220, 190, 85, 120, 95, 2, 1, 35,  
                  48, 72, 360, 410, 12, 20, 180, 240, 130, 145),
```

```
  AQI         = c(365, 110, 125, 180, 160, 72, 84, 290, 320, 240,  
                  175, 220, 58, 52, 270, 190, 145, 98, 60, 55)
```

```
)
```

Plot the climate and air quality analysis

```
ggplot(climate, aes(x = Temperature, y = AQI, size = Rainfall, colour = Humidity)) +
```

```
  geom_point(alpha = 0.85) +
```

```
  scale_colour_viridis_c(option = "C") +
```

```
  scale_size_continuous(range = c(3, 12)) +
```

```
  labs(
```

```
    title = "Indian Climate and Air Quality Analysis",
```

```
    x     = "Temperature (°C)",
```

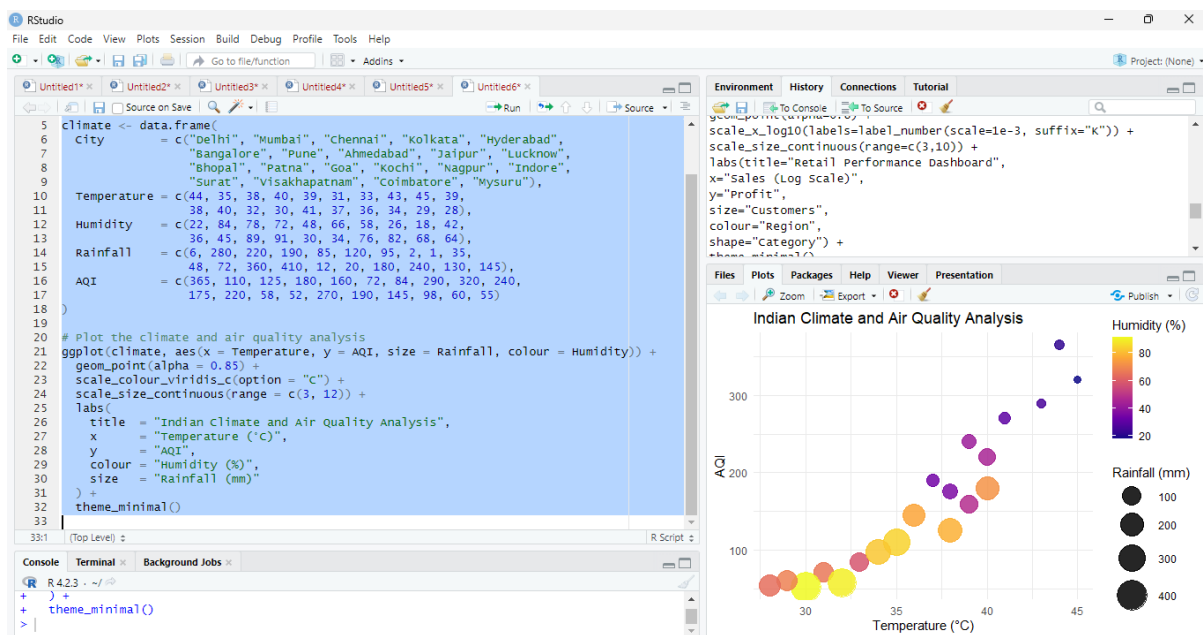
```
    y     = "AQI",
```

```
    colour = "Humidity (%)",
```

```
    size   = "Rainfall (mm)"
```

```
  ) +
```

```
  theme_minimal()
```



Design Justification

- **Temperature** on x-axis and **AQI** on y-axis reveal heat–pollution relationships.
- **Rainfall** encoded by point size highlights monsoon intensity.
- **Humidity** encoded by colour uses the **Viridis palette**, which is perceptually uniform and colour-blind friendly.

Interpretation

- Delhi and Jaipur show very high AQI and temperature.
- Kochi and Goa have high rainfall and humidity but low AQI.
- Cooler southern cities generally exhibit better air quality.

Question 4 – Nonlinear Transformation for Skewed Software Data

```
library(ggplot2)
```

```
library(scales)
```

```
# Create the software team data frame
```

```
software <- data.frame(
```

```
  Team      = paste("T", 1:20, sep = ""),
```

```
  LinesOfCode = c(12000, 18000, 25000, 35000, 48000,
```

```
                  62000, 81000, 96000, 120000, 150000,
```

```
                  185000, 240000, 310000, 390000, 480000,
```

```
                  620000, 810000, 1050000, 1350000, 1800000),
```

```
  Bugs      = c(35, 52, 60, 72, 88,
```

```
               110, 125, 142, 165, 188,
```

```
               205, 240, 280, 320, 365,
```

```
               420, 490, 560, 640, 720),
```

```
  Developers = c(5, 7, 9, 10, 12,
```

```
               14, 16, 18, 20, 22,
```

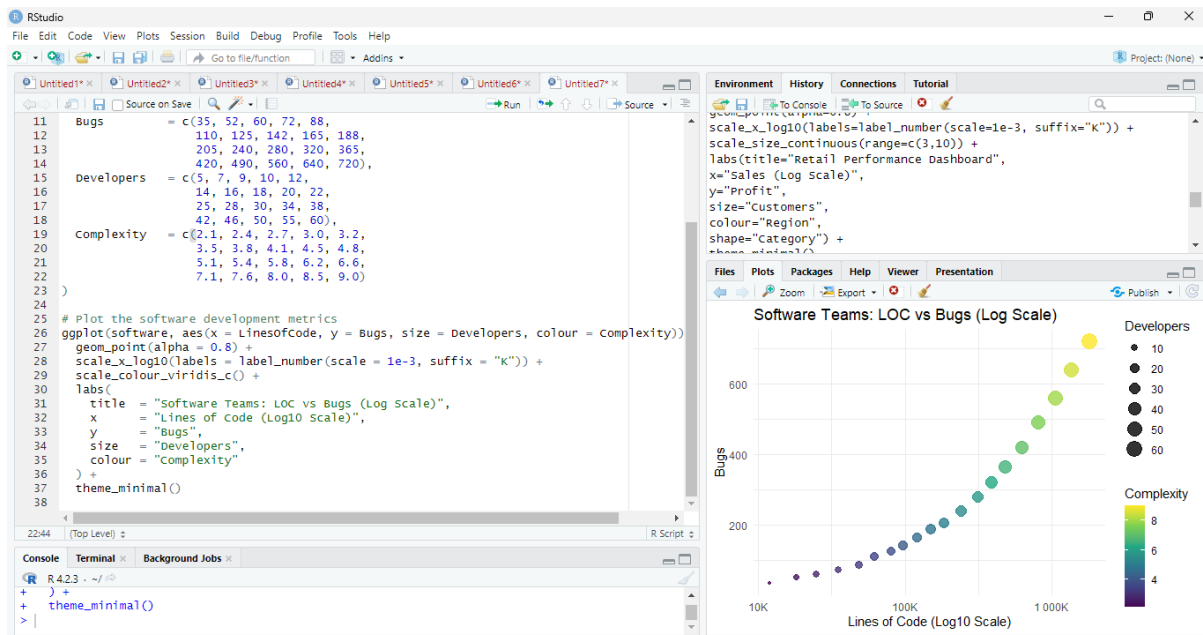


```

        25, 28, 30, 34, 38,
        42, 46, 50, 55, 60),
Complexity = c(2.1, 2.4, 2.7, 3.0, 3.2,
               3.5, 3.8, 4.1, 4.5, 4.8,
               5.1, 5.4, 5.8, 6.2, 6.6,
               7.1, 7.6, 8.0, 8.5, 9.0)
)

# Plot the software development metrics
ggplot(software, aes(x = LinesOfCode, y = Bugs, size = Developers, colour =
Complexity)) +
  geom_point(alpha = 0.8) +
  scale_x_log10(labels = label_number(scale = 1e-3, suffix = "K")) +
  scale_colour_viridis_c() +
  labs(
    title = "Software Teams: LOC vs Bugs (Log Scale)",
    x     = "Lines of Code (Log10 Scale)",
    y     = "Bugs",
    size  = "Developers",
    colour = "Complexity"
  ) +
  theme_minimal()

```



Justification

- Lines of Code ranges from **12,000 to 1,800,000**, creating severe right skew.
- A **log₁₀ transformation** compresses extreme values and spreads smaller values.

Improvement over Linear Scale

- Reveals relationships among small and medium teams that would otherwise cluster together.
- Makes growth patterns easier to compare.
- Improves readability and reduces visual dominance of very large projects.

Question 5 – Integrated Financial Analytics Visualization

```
library(ggplot2)
```

```
library(scales)
```

```
# Create the company metrics data frame
```

```
companies <- data.frame(  
  Company    = paste("C", 1:20, sep = ""),  
  Sector     = c("IT", "Banking", "Healthcare", "Retail", "Energy",  
                 "IT", "Banking", "Healthcare", "Retail", "Energy",  
                 "IT", "Banking", "Healthcare", "Retail", "Energy",  
                 "IT", "Banking", "Healthcare", "Retail", "Energy"),  
  Revenue    = c(420, 580, 720, 850, 980,  
                 1150, 1320, 1480, 1650, 1820,  
                 2050, 2280, 2550, 2840, 3150,  
                 3480, 3860, 4250, 4680, 5200),  
  ProfitMargin = c(8.2, 15.6, 11.4, 9.3, 18.5,  
                  20.4, 17.8, 14.2, 10.5, 22.1,  
                  24.5, 19.8, 16.7, 13.4, 26.2,  
                  28.1, 21.5, 18.3, 15.2, 30.4),  
  Employees   = c(1500, 3200, 2100, 4200, 2800,  
                  5100, 6800, 3900, 7200, 4500,  
                  8100, 9500, 6200, 10300, 7200,  
                  11800, 13200, 8600, 14500, 16000),  
  MarketCap   = c(8500, 12000, 9800, 7600, 14500,  
                  18200, 21500, 16800, 13200, 24500,  
                  28500, 32500, 24800, 19200, 38200,  
                  42500, 46800, 35600, 29800, 52000)  
)
```

```
# Plot integrated company performance multi-panel dashboard
```

```
ggplot(companies, aes(x = Revenue, y = ProfitMargin, size = Employees, colour = Sector)) +  
  geom_point(alpha = 0.8) +  
  scale_size_continuous(range = c(3, 12)) +
```

```

scale_colour_brewer(palette = "Set2") +

scale_x_continuous(labels = label_number(suffix = "M")) +

facet_wrap(~Sector) +

labs(

  title = "Integrated Company Performance Visualization",

  x     = "Revenue (Million)",

  y     = "Profit Margin (%)",

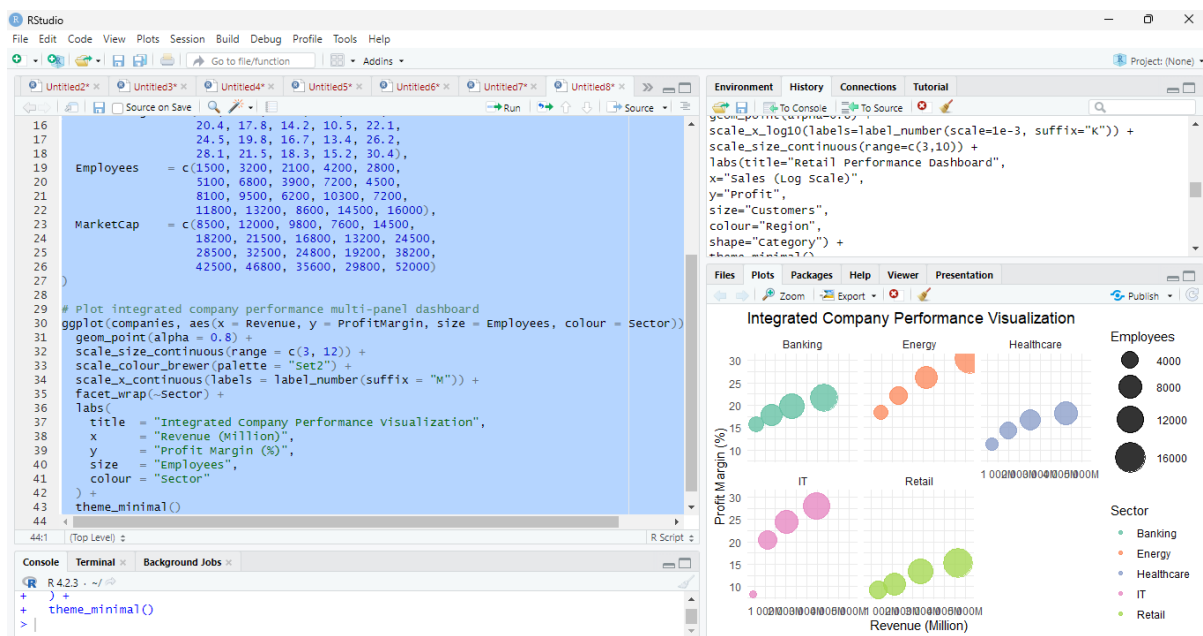
  size  = "Employees",

  colour = "Sector"

) +

theme_minimal()

```



Design Justification

- **Revenue** on x-axis and **Profit Margin** on y-axis capture scale and profitability.
- **Employees** mapped to size indicate organisational scale.
- **Sector** mapped to colour and faceting avoids overlap and improves comparison.
- **Set2 palette** provides balanced categorical colours with reduced perceptual bias.

Interpretation

- Energy companies show the highest profit margins.
- IT companies exhibit strong profitability with large revenues.
- Retail companies have larger employee counts but comparatively lower margins.
- Banking companies occupy a middle performance range.

Conclusion

The integrated bubble-facet visualization communicates five variables simultaneously while maintaining clarity, reducing clutter, and supporting sector-wise comparison for strategic financial analysis.